

Designing an Explainable AI-Based Social Interaction Support System for Children with Autism Spectrum Disorder

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Introduction & Background



ASD Prevalence

In the UK, approximately 1 in 57 children are on the autistic spectrum, equivalent to about 1.76% of the childhood population (University of Cambridge, 2021). Prevalence also varies by sex, with around 2.8% of boys and 0.65% of girls identified as autistic. The commonly cited NHS estimate of “1 in 100” is likely an underestimate, partly due to significant diagnostic backlogs and underdiagnosis, meaning the true prevalence may be higher than officially reported figures suggest.



Core Challenge

Facial emotion recognition (FER) is essential for social interaction, yet children with ASD often struggle in this area. Difficulties in interpreting facial expressions and nonverbal cues, along with challenges in back-and-forth conversation and reduced sharing of interests, limit peer relationships, educational engagement, and overall quality of life. These social communication challenges are further compounded by restricted and repetitive behaviours, including sensory sensitivities and rigid adherence to routines, making everyday social interaction and emotional connection more difficult.



Technology Gap

AI tools for ASD are often seen as “black-box” systems because they assign emotion labels without explaining their reasoning. This lack of transparency reduces trust for users, educators, and caregivers and makes it difficult to verify results. It also limits educational value, as children are not guided in understanding emotional cues. In addition, the opacity raises ethical concerns around accountability, bias, and the risk of misleading interpretations, particularly for vulnerable users.

The Black-Box Problem in ASD Support

1

Eroded Trust & Accountability

Therapists and caregivers cannot evaluate AI reliability without knowing the reasoning behind predictions. When systems fail, there is no way to assign accountability or diagnose errors.

2

No Educational Value

Labels like "Happy" or "Sad" teach nothing. Children with ASD need to understand which facial feature signals the emotion upturned mouth, crinkled eyes, furrowed brows.

3

Ethical Risks & Harm

Unexplained AI can perpetuate demographic bias, undermine user autonomy, and risk stigmatising neurodiverse children through inappropriate or unexplained feedback.

Literature Review — Existing Solutions & Gaps

What Exists

Social Robots

NAO, KASPAR — structured interaction but costly and rigid

Computer Apps

FaceSay, Mind Reading ; FER training but no real-time feedback

Deep Learning FER

CNNs reach >73% on FER2013 ; but fully black-box systems

XAI Methods

Grad-CAM, LIME : powerful but never applied to ASD assistive tech

Critical Research Gaps

- No XAI explanations integrated into any ASD assistive technology
- Poor in-the-wild FER performance (ExpW is far harder than FER2013)
- Documented demographic bias unaddressed in deployed FER systems
- No real-time, user-tested, explainable system exists for ASD context
- Ethical frameworks for AI facial analysis in ASD are underdeveloped

Key refs: Baron-Cohen (2004), Selvaraju et al. (2017) Grad-CAM, Buolamwini & Gebru (2018), Rudin (2019), Wang & Vasconcelos (2019)

Aim, Objectives & Research Questions

Aim: Design, develop and evaluate an XAI-based social interaction support system for children with ASD that prioritises transparency, educational value and ethical responsibility.

04 Objectives

- 1 Review ASD, FER and XAI literature to establish design foundations
- 2 Build and evaluate 4 custom CNN architectures on the ExpW dataset
- 3 Implement Grad-CAM to generate child-friendly visual explanations
- 4 Derive ethical design guidelines for AI systems for neurodiverse users

Research Questions

RQ1 Accuracy and Computational Feasibility

To what extent can custom CNN architecture trained on an in-the-wild FER dataset achieve accuracy levels sufficiently to motivate further development as an ASD support tool?

RQ2 Grad-CAM plausibility and explanation Quality

Can Grad-CAM-based visual explanation generate anatomically plausible saliency maps that, combined with rule-based natural language generation, produce child-appropriate explanation of facial emotion predictions?

RQ3 Technical limitation and Future research Agenda

What are the principal technical limitations of the current approach, and what future research steps are required to progress from computational feasibility towards a validated, ethically deployed system?

Research Methodology — Sequential Mixed-Methods Design

⚠ Important: No direct interaction with autistic children or caregivers was conducted in this research. Phases 1 & 3 are literature-grounded proposed frameworks only.

Phase 1

Requirements Elicitation

- Literature review
- Stakeholder analysis
- Design principles derivation
- (Proposed – not conducted)

Phase 2

Technical Development

- ExpW dataset – 59,442 images
- 4 Custom CNN architectures
- Grad-CAM XAI module
- Full performance evaluation

Phase 3

Usability & Trust Evaluation

- SUS usability testing
- Trust & learning outcomes
- Qualitative user experience
- (Proposed – not conducted)

★ EMPIRICAL CORE OF THIS RESEARCH

Dataset Collection & Implementation Pipeline

106,962

Raw Images
(ExpW Dataset)

59,442

Valid Labelled
Images

47,520

Skipped
(No match / corrupt)

4

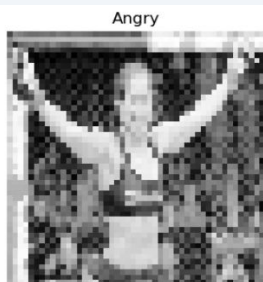
Emotion Classes
Angry/Happy/Sad/Neutral

Dataset Split (Stratified, seed=42)

Split	Samples	Angry	Happy	Sad	Neutral
Training (60%)	35,665	4,908	2,612	4,857	512
Validation (20%)	11,888	1,636	871	1,619	171
Test (20%)	11,889	4,858	2,453	4,565	13

⚠️ **Neutral: only 696 total samples — severe class imbalance**

Sample Dataset Images - ExpW (48x48 grayscale, showing Angry, Happy and Sad)



Preprocessing Pipeline



Collect:

ExpW via Kaggle API — 106,962 images across 7 expression folders



Label Map:

7 classes → 4 target emotions; invalid filenames discarded



Preprocess:

Greyscale → resize 48x48 → histogram EQ → Gaussian blur (3x3) → normalise [0,1]



Split:

Stratified 60/20/20 split; seed=42 for full reproducibility



Augment:

Random flip, ±10° rotation, ±10% zoom applied to training set only



Pipeline:

tf.data with AUTOTUNE; batch=64; prefetch for GPU efficiency

Custom CNN Architectures — Design Overview

Model	Conv Blocks	Filter Depth	Dense Head	Total Params	Input Shape
CNN-v1	3 Blocks × 2 Conv	32 → 64 → 128	Dense(128)	305,252	48×48×1
CNN-v2	4 Blocks × 2 Conv	32 → 64 → 128 → 256	Dense(256)	1,242,340	48×48×1
CNN-v3	3 Blocks × 1 Conv	32 → 64 → 128	Dense(128)	110,596	48×48×1
CNN-v4	3 Blocks × 2 Conv	32 → 64 → 128	Dense(128)	315,748	48×48×1

Common Design Features (All 4 Models)



Each Conv block:

includes 2D Convolution → Batch Normalisation → ReLU → MaxPool(2×2) → Dropout(0.25)



Training:

Adam (lr=0.001) | Categorical cross-entropy | EarlyStopping(patience=10) | ReduceLROnPlateau



Classification head:

Global Average Pooling → Dense(ReLU) → Dropout(0.5) → Dense(4, Softmax)



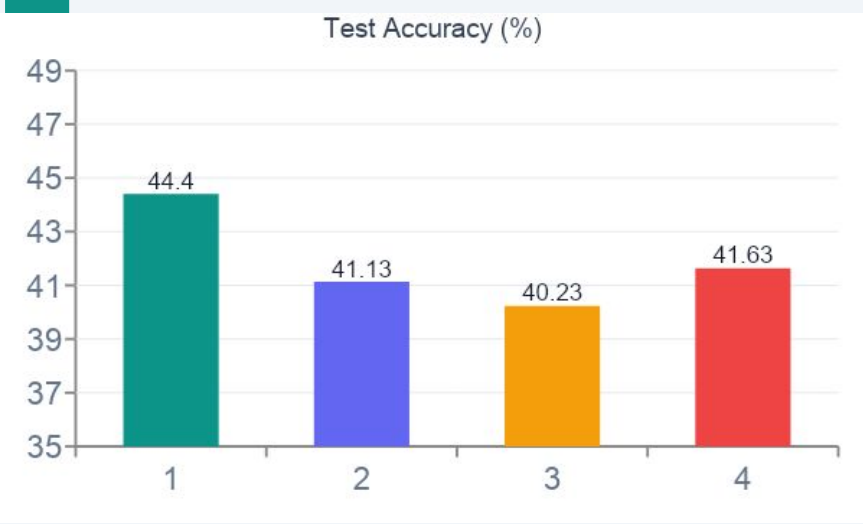
Augmentation:

RandomFlip("horizontal") | RandomRotation(0.1) | RandomZoom(0.1) applied in-graph

Custom CNN Results — Full Performance Summary

Model	Test Accuracy	Precision	Recall	Test Loss	Parameters
CNN-v1 ★	44.40%	0.5495	0.0840	1.0554	305,252
CNN-v2	41.13%	0.4297	0.0611	1.0732	1,242,340
CNN-v3	40.23%	0.4202	0.1272	1.2086	110,596
CNN-v4	41.63%	0.6486	0.0020	1.0702	315,748

★ CNN-v1 is the best-performing custom model: highest accuracy (44.40%) with only 305K parameters — efficient and effective



Per-Class Performance — CNN-v1 (Best Model)

Class	Precision	Recall	F1-Score	Support
Angry	0.45	0.72	0.55	4,858
Happy	0.37	0.01	0.02	2,453
Sad	0.44	0.39	0.41	4,565
Neutral	0.00	0.00	0.00	13

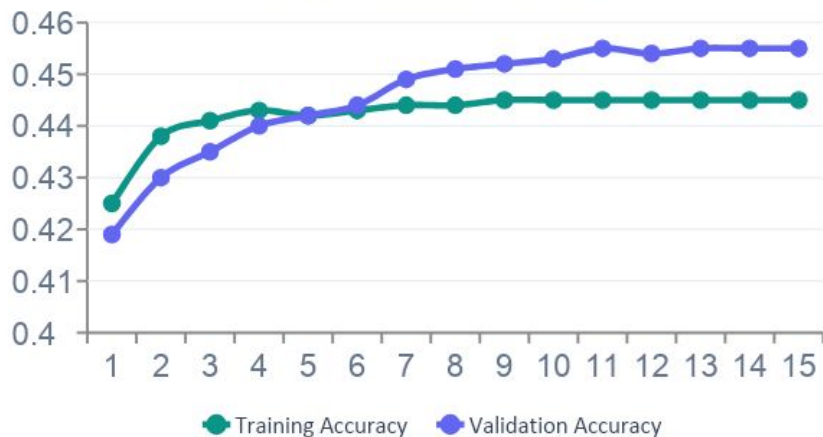
⚠ Key Issue — Class Imbalance:

Neutral (13 test samples) and Happy (2,453) are severely under-represented vs. Angry (4,858), resulting in near-zero recall for Neutral and Happy - a critical limitation to address next.

This issue was further compounded by limited computational resources, as the available GPU was slow and constrained the efficiency of training the custom models.

Deep Dive — CNN-v1 (Best Model) Training Curves

Training vs Validation Accuracy



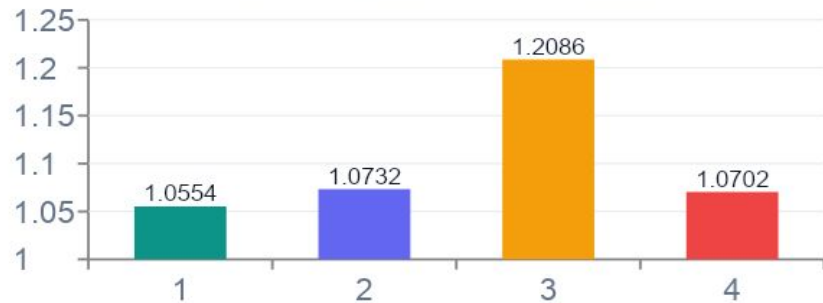
Training vs Validation Loss



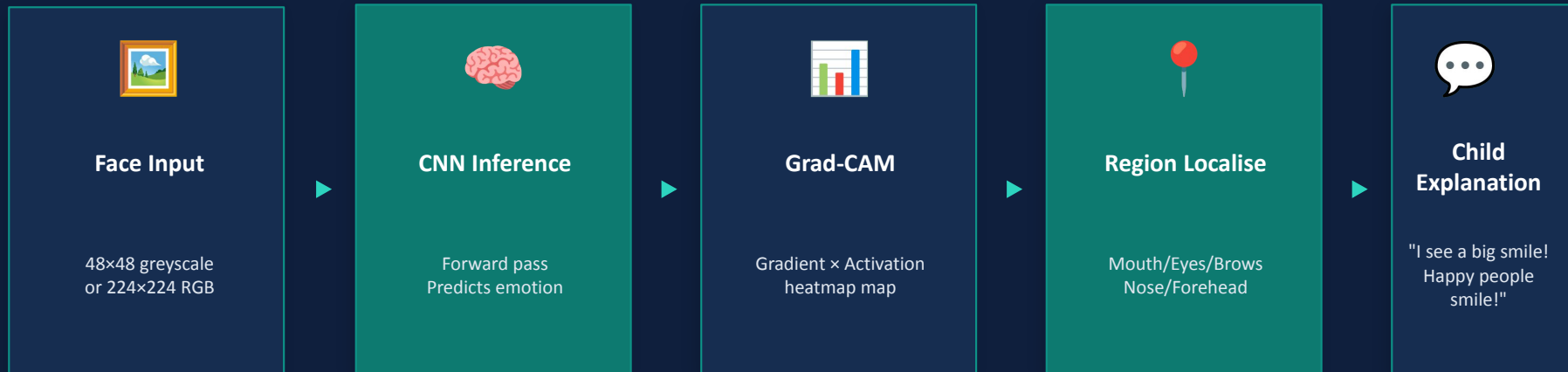
Precision vs Recall — All 4 Models



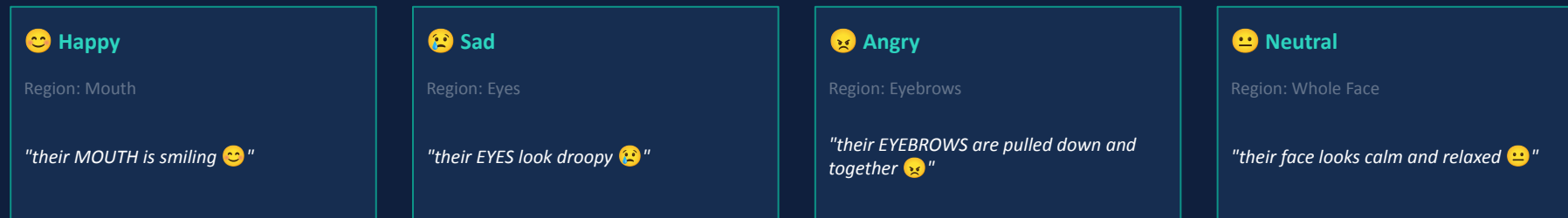
Test Loss Comparison (lower is better)



Grad-CAM Explainability — How It Works



Sample Explanations Generated by the Module:



Project Progress & Next Steps

✓ Completed

- Comprehensive literature review (ASD, FER, XAI)
- ExpW dataset collection, cleaning and label mapping
- 4 custom CNN architectures designed and trained
- CNN-v1: 44.40% test accuracy (best custom model)
- Grad-CAM explainability module implemented
- Child-friendly natural language explanation system
- 6 facial regions defined for explanation targeting
- Full evaluation: accuracy, precision, recall, loss

➡ Up Next

Transfer Learning Models

Fine-tune VGG16, VGG19, ResNet50 and ResNet152 on ExpW; compare all against custom CNN baselines

Class Imbalance Mitigation

Explore class-weighted loss functions and oversampling strategies to fix near-zero recall for Neutral and Happy

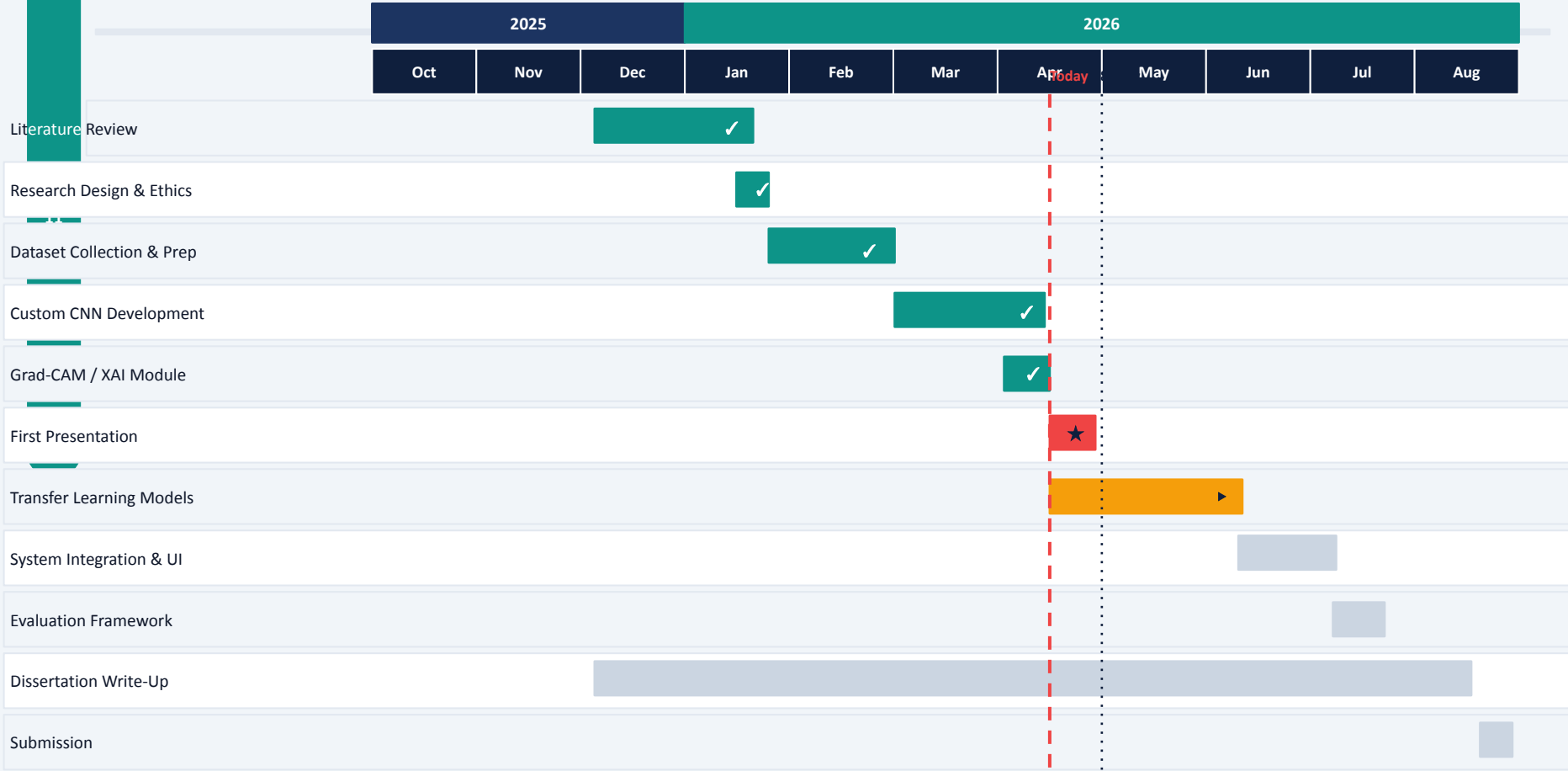
System Integration & UI

Connect XAI module with real-time video pipeline; build accessible, ASD-appropriate user interface

Full Dissertation Write-Up

Complete all chapters incorporating final transfer learning results and comparative analysis

Project Timeline — Gantt Chart & Key Milestones



Key Milestones & Deliverables

Milestone	Target Date	Key Deliverable	Status
M1 — Literature Review Complete	Jan 2026	Chapter 2 draft	✓ Done
M2 — Research Design Finalised	Jan 2026	Chapter 3 methodology draft	✓ Done
M3 — Dataset Ready	Feb 2026	59,442 labelled ExpW images	✓ Done
M4 — Custom CNNs Trained	Apr 2026	4 model results + full analysis	✓ Done
M5 — Grad-CAM XAI Module Built	Apr 2026	Explainability	✓ Done
M6 — First Presentation	08 May 2026	This presentation (PDF)	 Due
M7 — Transfer Learning Models	Jun 2026	VGG16/19, ResNet50/152 results	 Next
M8 — System Integration & UI	Jul 2026	Integrated XAI-SISS prototype + explanation UI	 Planned
M9 — Dissertation Submitted	Aug 2026	Final dissertation (PDF on Canvas)	 Planned

Overall Progress

5 / 9

56% Complete (5 of 9 milestones)

Thank You

Questions & Discussion

59K

Training
Images

4

CNN Models
Trained

44.4%

Best Custom
Accuracy

14.7

Frames/sec
Real-Time

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Submission: August 2026